# LogNormal

## Parametrisation

The LogNormal has density

$$f(y) = \frac{1}{y\sqrt{2\pi}}\sqrt{\tau}\exp\left(-\frac{1}{2}\tau(\log y - \mu)^2\right), \qquad y > 0$$

where

$\tau > 0$ is the precision parameter,

$\mu$ is the mean parameter.

## Link-function

The parameter $\mu$ is linked to the linear predictor as:

$$\eta = \mu$$

## Hyperparameters

The $\tau$ parameter is represented as

$$\theta = \log\tau$$

and the prior is defined on $\theta$.

## Specification

- `family="lognormal"` for regression models and `family="lognormalsurv"` for survival models.
- Required arguments: $y$. Given in a format by using `inla.surv()` function for `"lognormal.surv"`

### Hyperparameter spesification and default values

**lognormal**

**doc** `The log-Normal likelihood`
**hyper**
**theta**
**hyperid** `77101`
**name** `log precision`
**short.name** `prec`
**output.name** `Precision for the lognormal observations`
**output.name.intern** `Log precision for the lognormal observations`
**initial** `0`
**fixed** `FALSE`
**prior** `loggamma`
**param** `1 5e-05`
**to.theta** `function(x) log(x)`
**from.theta** `function(x) exp(x)`
**survival** `FALSE`
**discrete** `FALSE`
**link** `default identity`
**pdf** `lognormal`

**lognormalsurv**

**doc** The log-Normal likelihood (survival)

**hyper**

**theta1**

**hyperid** 78001

**name** log precision

**short.name** prec

**output.name** Precision for the lognormalsurv observations

**output.name.intern** Log precision for the lognormalsurv observations

**initial** 0

**fixed** FALSE

**prior** loggamma

**param** 1 5e-05

**to.theta** function(x) log(x)

**from.theta** function(x) exp(x)

**theta2**

**hyperid** 78002

**name** beta1

**short.name** beta1

**output.name** beta1 for logNormal-Cure

**output.name.intern** beta1 for logNormal-Cure

**initial** -7

**fixed** FALSE

**prior** normal

**param** -4 100

**to.theta** function(x) x

**from.theta** function(x) x

**theta3**

**hyperid** 78003

**name** beta2

**short.name** beta2

**output.name** beta2 for logNormal-Cure

**output.name.intern** beta2 for logNormal-Cure

**initial** 0

**fixed** FALSE

**prior** normal

**param** 0 100

**to.theta** function(x) x

**from.theta** function(x) x

**theta4**

**hyperid** 78004

**name** beta3

**short.name** beta3

**output.name** beta3 for logNormal-Cure

**output.name.intern** `beta3 for logNormal-Cure`
**initial** `0`
**fixed** `FALSE`
**prior** `normal`
**param** `0 100`
**to.theta** `function(x) x`
**from.theta** `function(x) x`

**theta5**
**hyperid** `78005`
**name** `beta4`
**short.name** `beta4`
**output.name** `beta4 for logNormal-Cure`
**output.name.intern** `beta4 for logNormal-Cure`
**initial** `0`
**fixed** `FALSE`
**prior** `normal`
**param** `0 100`
**to.theta** `function(x) x`
**from.theta** `function(x) x`

**theta6**
**hyperid** `78006`
**name** `beta5`
**short.name** `beta5`
**output.name** `beta5 for logNormal-Cure`
**output.name.intern** `beta5 for logNormal-Cure`
**initial** `0`
**fixed** `FALSE`
**prior** `normal`
**param** `0 100`
**to.theta** `function(x) x`
**from.theta** `function(x) x`

**theta7**
**hyperid** `78007`
**name** `beta6`
**short.name** `beta6`
**output.name** `beta6 for logNormal-Cure`
**output.name.intern** `beta6 for logNormal-Cure`
**initial** `0`
**fixed** `FALSE`
**prior** `normal`
**param** `0 100`
**to.theta** `function(x) x`
**from.theta** `function(x) x`

**theta8**
**hyperid** `78008`

**name** beta7
**short.name** beta7
**output.name** beta7 for logNormal-Cure
**output.name.intern** beta7 for logNormal-Cure
**initial** 0
**fixed** FALSE
**prior** normal
**param** 0 100
**to.theta** function(x) x
**from.theta** function(x) x

**theta9**
**hyperid** 78009
**name** beta8
**short.name** beta8
**output.name** beta8 for logNormal-Cure
**output.name.intern** beta8 for logNormal-Cure
**initial** 0
**fixed** FALSE
**prior** normal
**param** 0 100
**to.theta** function(x) x
**from.theta** function(x) x

**theta10**
**hyperid** 78010
**name** beta9
**short.name** beta9
**output.name** beta9 for logNormal-Cure
**output.name.intern** beta9 for logNormal-Cure
**initial** 0
**fixed** FALSE
**prior** normal
**param** 0 100
**to.theta** function(x) x
**from.theta** function(x) x

**theta11**
**hyperid** 78011
**name** beta10
**short.name** beta10
**output.name** beta10 for logNormal-Cure
**output.name.intern** beta10 for logNormal-Cure
**initial** 0
**fixed** FALSE
**prior** normal
**param** 0 100
**to.theta** function(x) x

**from.theta** `function(x) x`

**survival** `TRUE`

**discrete** `FALSE`

**link** `default identity`

**pdf** `lognormal`

## Example

In the following example we estimate the parameters in a simulated case

```
n = 300
x = c(scale(runif(n)))
eta = 1+2.2*x
y = exp(rnorm(n, mean = eta,  sd = 1))
data = list(y=y, event=rep(1, n), x=x)
formula = inla.surv(y, event) ~ 1 + x
r=inla(formula, family ="lognormalsurv", data=data)
summary(r)

data = data.frame(y, x)
formula = y  ~ 1 + x
r=inla(formula, family ="lognormal", data=data)
summary(r)
```

## Notes

- lognormalsurv can be used for right censored, left censored, interval censored data. A general framework to represent time is given by `inla.surv`. If the observed times $y$ are large/huge, then this can cause numerical overflow, and if you encounter this problem, try to scale the observatios, like `time = time / max(time)`.